Chapter 17 Binomial Distributions - Homework Answers

Chapter revision | Warm-up

Table values are from the standard normal table at the end of the Revision section.

1 Exercise 16D Q9

a $P(Z \leq 1.3) = \Phi(1.3) \doteq 0.9032$

b $P(Z = 2.4) = 0$, because the probability of any single value is zero.

c $P(Z > 0.4) = 1 - \Phi(0.4) \doteq 1 - 0.6554 = 0.3446$

d $P(Z \leq -0.8) = P(Z \geq 0.8) = 1 - \Phi(0.8) \doteq 1 - 0.7881 = 0.2119$

e $P(1.1 \leq Z \leq 1.5) = \Phi(1.5) - \Phi(1.1) \doteq 0.9332 - 0.8643 = 0.0689$

f $P(-1.3 \leq Z \leq 2.2) = \Phi(2.2) - \big(1 - \Phi(1.3)\big) \doteq 0.9861 - 0.0968 = 0.8893$

2 Exercise 16D Q12

a $P(Z \leq 1) \doteq 50\% + 34\% = 84\%$

b $P(Z < -1) \doteq \frac{1}{2}(100\% - 68\%) = 16\%$

c $P(0 \leq Z \leq 3) \doteq \frac{1}{2} \times 99.7\% = 49.85\%$

d $P(-3 < Z \leq -2) \doteq \frac{1}{2}(99.7\% - 95\%) = 2.35\%$

e $P(-2 \leq Z < 1) \doteq 47.5\% + 34\% = 81.5\%$

3 Exercise 16D Q14

a $\Phi(a) = 0.7257$, so $a = 0.6$.

b $P(Z < -a) = 1 - \Phi(a) = 0.1151$, so $\Phi(a) = 0.8849$ and $a = 1.2$.

c The probability is less than 0.5, so a is negative. $\Phi(-a) = 1 - 0.2119 = 0.7881$, so $-a = 0.8$ and $a = -0.8$.

d $P(-a \leq Z < a) = 2\Phi(a) - 1 = 0.7286$, so $\Phi(a) = 0.8643$ and $a = 1.1$.

4 Exercise 16E Q6

a $z = \dfrac{3-5}{0.8} = -2.5$ and $z = \dfrac{7-5}{0.8} = 2.5$, so
$P(3 \leq X \leq 7) = P(-2.5 \leq Z \leq 2.5) = 2\Phi(2.5) - 1 \doteq 2 \times 0.9938 - 1 = 0.9876$.

b $z = \dfrac{8-12}{5} = -0.8$, so
$P(X \leq 8) = P(Z \leq -0.8) = 1 - \Phi(0.8) \doteq 1 - 0.7881 = 0.2119$.

c $z = \dfrac{-39-0}{30} = -1.3$, so
$P(X \geq -39) = P(Z \geq -1.3) = \Phi(1.3) \doteq 0.9032$.

d $z = \dfrac{3-8}{2} = -2.5$ and $z = \dfrac{5-8}{2} = -1.5$, so
$P(3 < X \leq 5) = P(-2.5 < Z \leq -1.5) = \Phi(2.5) - \Phi(1.5) \doteq 0.9938 - 0.9332 = 0.0606.$

5 Exercise 16F Q2

a $z = \dfrac{95-68}{9} = 3$. By the empirical rule, $P(Z > 3) \doteq \frac{1}{2}(100\% - 99.7\%) = 0.15\%$, so about $2000 \times 0.0015 = 3$ students.

b $z = \dfrac{50-68}{9} = -2$. $P(Z < -2) \doteq \frac{1}{2}(100\% - 95\%) = 2.5\%$, so about $2000 \times 0.025 = 50$ students.

c $z = \dfrac{59-68}{9} = -1$ and $z = \dfrac{86-68}{9} = 2$. $P(-1 < Z < 2) \doteq 34\% + 47.5\% = 81.5\%$, so about $2000 \times 0.815 = 1630$ students.

6 Exercise 16F Q7

a From the table, $\Phi(1.28) \doteq 0.8997$ and $\Phi(1.29) \doteq 0.9015$. The closer value to 0.9 is 0.8997, so $z \doteq 1.28$.

b $x = \mu + \sigma z = 176 + 1.28 \times 7.5 = 185.6$, so the doorway must be about 186 cm high.

c i From the table, $\Phi(1.64) \doteq 0.9495$ and $\Phi(1.65) \doteq 0.9505$. The value 0.95 lies halfway between them, so $z \doteq 1.645$, and 1.65 is a reasonable estimate (1.645 rounds to 1.65). Interpolating between $\Phi(1.6) \doteq 0.9452$ and $\Phi(1.7) \doteq 0.9554$ also gives about 1.65.

ii $x = 185 + 1.65 \times 7.5 = 197.375$, so the door must be about 197 cm high.

Chapter 17 Binomial Distributions - Homework Answers

Lesson 17A | Binomial probability

Lesson Examples

C1 **Tutor demo.**

a (i) Yes. Success is 'an even number', with $p = \frac{3}{6} = \frac{1}{2}$.

(ii) No. There are six outcomes, not two.

(iii) Yes. Success is 'a heart', with $p = \frac{13}{52} = \frac{1}{4}$.

b (i) Yes, with $n = 10$ and $p = \frac{1}{2}$.

(ii) No. The number of stages is not fixed in advance.

(iii) No. The cards are not replaced, so the probability of a heart changes from draw to draw, and the stages are not independent.

Your turn.

a (i) Yes. Success is 'lands on 5', with $p = \frac{1}{5}$.

(ii) No. A height is not one of just two outcomes.

(iii) Yes. Success is 'a number greater than 4', that is, 5 or 6, with $p = \frac{2}{6} = \frac{1}{3}$.

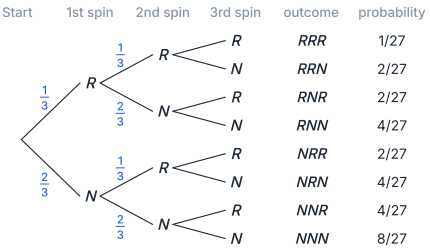
b (i) Yes, with $n = 8$ and $p = \frac{1}{6}$.

(ii) No. The balls are not replaced, so the stages are not independent.

(iii) No. X is a total of the numbers, not a number of successes.

C2 **Tutor demo.** Let R be 'lands on red' and N be 'does not land on red', so $p = \frac{1}{3}$ and $q = \frac{2}{3}$.

a Each R branch has probability $\frac{1}{3}$ and each N branch has probability $\frac{2}{3}$.

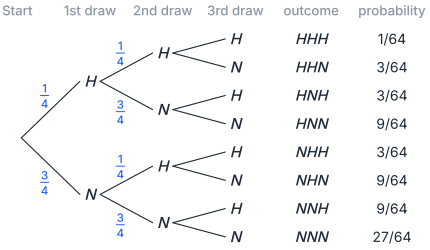


b $P(X = 0) = {}^3C_0\left(\frac{1}{3}\right)^0\left(\frac{2}{3}\right)^3 = \frac{8}{27}$, $P(X = 1) = {}^3C_1\left(\frac{1}{3}\right)^1\left(\frac{2}{3}\right)^2 = \frac{12}{27}$,
$P(X = 2) = {}^3C_2\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^1 = \frac{6}{27}$, $P(X = 3) = {}^3C_3\left(\frac{1}{3}\right)^3\left(\frac{2}{3}\right)^0 = \frac{1}{27}$

c $\left(\frac{1}{3} + \frac{2}{3}\right)^3 = {}^3C_0\left(\frac{1}{3}\right)^0\left(\frac{2}{3}\right)^3 + {}^3C_1\left(\frac{1}{3}\right)^1\left(\frac{2}{3}\right)^2 + {}^3C_2\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^1 + {}^3C_3\left(\frac{1}{3}\right)^3\left(\frac{2}{3}\right)^0$
$= \frac{8}{27} + \frac{12}{27} + \frac{6}{27} + \frac{1}{27} = 1$. These are the four probabilities in part **b**.

Your turn. Let H be 'a heart' and N be 'not a heart', so $p = \frac{1}{4}$ and $q = \frac{3}{4}$.

a Each H branch has probability $\frac{1}{4}$ and each N branch has probability $\frac{3}{4}$.



b $P(X = 0) = {}^3C_0\left(\frac{1}{4}\right)^0\left(\frac{3}{4}\right)^3 = \frac{27}{64}$, $P(X = 1) = {}^3C_1\left(\frac{1}{4}\right)^1\left(\frac{3}{4}\right)^2 = \frac{27}{64}$,
$P(X = 2) = {}^3C_2\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^1 = \frac{9}{64}$, $P(X = 3) = {}^3C_3\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^0 = \frac{1}{64}$

c $\left(\frac{1}{4} + \frac{3}{4}\right)^3 = {}^3C_0\left(\frac{1}{4}\right)^0\left(\frac{3}{4}\right)^3 + {}^3C_1\left(\frac{1}{4}\right)^1\left(\frac{3}{4}\right)^2 + {}^3C_2\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^1 + {}^3C_3\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^0$
$= \frac{27}{64} + \frac{27}{64} + \frac{9}{64} + \frac{1}{64} = 1$. These are the four probabilities in part **b**.

C3 **Tutor demo.** A number less than 3 is 1 or 2, so $p = \frac{1}{3}$ and $q = \frac{2}{3}$, with $n = 7$.

a $^7C_2 \times \left(\frac{1}{3}\right)^2 \times \left(\frac{2}{3}\right)^5 = \dfrac{21 \times 2^5}{3^7} = \dfrac{672}{3^7}$

b $^7C_1 \times \left(\frac{1}{3}\right)^1 \times \left(\frac{2}{3}\right)^6 = \dfrac{7 \times 2^6}{3^7} = \dfrac{448}{3^7}$

c $1 - \left(\frac{2}{3}\right)^7 = 1 - \dfrac{128}{3^7} = \dfrac{2059}{3^7}$

d $^7C_6 \times \left(\frac{1}{3}\right)^6 \times \left(\frac{2}{3}\right)^1 + {}^7C_7 \times \left(\frac{1}{3}\right)^7 \times \left(\frac{2}{3}\right)^0 = \dfrac{7 \times 2 + 1}{3^7} = \dfrac{15}{3^7}$

Your turn. There are 13 hearts in the pack, so $p = \frac{1}{4}$ and $q = \frac{3}{4}$, with $n = 5$.

a $^5C_2 \times \left(\frac{1}{4}\right)^2 \times \left(\frac{3}{4}\right)^3 = \dfrac{10 \times 3^3}{4^5} = \dfrac{270}{4^5}$

b $^5C_1 \times \left(\frac{1}{4}\right)^1 \times \left(\frac{3}{4}\right)^4 = \dfrac{5 \times 3^4}{4^5} = \dfrac{405}{4^5}$

c $1 - \left(\frac{3}{4}\right)^5 = 1 - \dfrac{243}{4^5} = \dfrac{781}{4^5}$

d $^5C_4 \times \left(\frac{1}{4}\right)^4 \times \left(\frac{3}{4}\right)^1 + {}^5C_5 \times \left(\frac{1}{4}\right)^5 \times \left(\frac{3}{4}\right)^0 = \dfrac{5 \times 3 + 1}{4^5} = \dfrac{16}{4^5}$

C4 **Tutor demo.** Take $p = q = \frac{1}{2}$ and $n = 20$.

a $P(\text{10 heads}) = {}^{20}C_{10} \times \left(\frac{1}{2}\right)^{20} = \dfrac{184\,756}{1\,048\,576} \doteq 0.1762$

b $P(\text{12 heads}) = {}^{20}C_{12} \times \left(\frac{1}{2}\right)^{20} = \dfrac{125\,970}{1\,048\,576} \doteq 0.1201$

Your turn. Take $p = q = \frac{1}{2}$ and $n = 30$.

a $P(\text{15 heads}) = {}^{30}C_{15} \times \left(\frac{1}{2}\right)^{30} \doteq 0.1445$

b $P(\text{18 heads}) = {}^{30}C_{18} \times \left(\frac{1}{2}\right)^{30} \doteq 0.08055$

C5 **Tutor demo.** Let X be the number of cracked eggs in a tray, with $p = 0.02$, $q = 0.98$ and $n = 30$.

a $P(X = 0) = 0.98^{30} \doteq 0.545$

b $P(X \geq 2) = 1 - \left(0.98^{30} + {}^{30}C_1 \times 0.02^1 \times 0.98^{29}\right) \doteq 0.121$

c For n eggs, $P(\text{at least one cracked}) = 1 - 0.98^n$. We need $1 - 0.98^n > 0.9$, so $0.98^n < 0.1$. Taking logarithms and dividing by $\ln 0.98 < 0$ reverses the sign: $n > \dfrac{\ln 0.1}{\ln 0.98} \doteq 113.97$. So 114 eggs must be checked. (Check: $0.98^{113} \doteq 0.1020$ and $0.98^{114} \doteq 0.0999$.)

Your turn. Let X be the number of seeds in a packet that fail, with $p = 0.05$, $q = 0.95$ and $n = 40$.

a $P(X = 0) = 0.95^{40} \doteq 0.129$

b $P(X \geq 2) = 1 - \left(0.95^{40} + {}^{40}C_1 \times 0.05^1 \times 0.95^{39}\right) \doteq 0.601$

c We need $1 - 0.95^n > 0.95$, so $0.95^n < 0.05$, and $n > \dfrac{\ln 0.05}{\ln 0.95} \doteq 58.40$. So 59 seeds must be planted. (Check: $0.95^{58} \doteq 0.0510$ and $0.95^{59} \doteq 0.0485$.)

C6 **Tutor demo.** In one hour the machines make 100 bolts, 60 by A and 40 by B. For a bolt chosen at random,

$p = P(\text{A made it and it is faulty}) + P(\text{B made it and it is faulty}) = \frac{60}{100} \times \frac{3}{100} + \frac{40}{100} \times \frac{5}{100} = 0.038$, so $q = 0.962$.

Let X be the number of faulty bolts in a box of 20. Then $P(X \leq 1) = 0.962^{20} + {}^{20}C_1 \times 0.962^{19} \times 0.038 \doteq 0.825$.

Your turn. In one hour the ovens bake 100 loaves, 40 in Oven 1 and 60 in Oven 2. For a loaf chosen at random,

$p = \frac{40}{100} \times \frac{5}{100} + \frac{60}{100} \times \frac{10}{100} = 0.08$, so $q = 0.92$.

Let X be the number of burnt loaves in a crate of 12. Then $P(X \leq 1) = 0.92^{12} + {}^{12}C_1 \times 0.92^{11} \times 0.08 \doteq 0.751$.

Textbook Homework

1 a $\frac{1}{32}$ b $\frac{5}{16}$ c $\frac{5}{32}$ d $\frac{31}{32}$

2 $^6C_2\left(\frac{4}{5}\right)^4\left(\frac{1}{5}\right)^2$

3 a Because of the large size of the roll and the large number of males on the roll, the probability is fairly constant across the selection.

b $(0.65)^{12}$

4 a $^{12}C_3\left(\frac{5}{6}\right)^9\left(\frac{1}{6}\right)^3$ b $^{12}C_8\left(\frac{5}{6}\right)^4\left(\frac{1}{6}\right)^8$

c $^{12}C_{10}\left(\frac{5}{6}\right)^2\left(\frac{1}{6}\right)^{10} + {}^{12}C_{11}\left(\frac{5}{6}\right)^1\left(\frac{1}{6}\right)^{11} + \left(\frac{1}{6}\right)^{12}$

5 **a** 0.2009 **b** 0.7368 **c** 0.2632

6 **a** $\left(\frac{9}{10}\right)^{20} + {}^{20}C_1\left(\frac{9}{10}\right)^{19}\left(\frac{1}{10}\right) + {}^{20}C_2\left(\frac{9}{10}\right)^{18}\left(\frac{1}{10}\right)^2$ **b** $1 - \left(\frac{9}{10}\right)^{20}$

7 **a** $0.91^{10} + {}^{10}C_1 \times 0.91^9 \times 0.09 + {}^{10}C_2 \times 0.91^8 \times 0.09^2$
b $1 - 0.91^{10} - {}^{10}C_1 \times 0.91^9 \times 0.09$

8 **a** HHTT, HTHT, THHT, HTTH, THTH, TTHH
b There are 6 cases in the list. The probability that she wins is $\frac{6}{2^4} = \frac{3}{8}$.
c Each winning result is an ordering of two Hs and two Ts, as in the word HHTT. There are $\frac{4!}{2!\times 2!} = {}^4C_2 = 6$ arrangements, which agrees with part **b**.
d The number of ways of choosing the two coins that are to be heads up is ${}^4C_2$.

9 0.593

10 ${}^{31}C_3 \times 0.95^{28} \times 0.05^3$

11 **a** (i) 0.10764 (ii) 0.11372 **b** 0.78549

12 **a** 17 **b** 7

13 **a** (i) 0.487 (ii) 0.031
b First, we have assumed that boys and girls are equally likely. Secondly, we have assumed that in any one family, the events 'having a boy' and 'having a girl' are independent. Thirdly, we have assumed that every child is a boy or a girl.

14 **a** The argument is invalid. Normally, mathematics books are grouped together, so that once the shelf is chosen, one would expect all or none of the books to be mathematics books, thus the five stages are not independent events. The result would be true if the books were each chosen at random from the library.
b The argument is invalid. People in a particular neighbourhood tend to vote more similarly than the population at large, so the four events are not independent. This method also oversamples small streets, which may introduce an additional bias.

15 **a** 0.0346 (exactly 0.03456) **b** 0.6826 (exactly 0.68256)

16 **a** 0.409600 **b** 0.001126 **c** 0.000869

17 **a** 0.0060 **b** 0.0303

18 **a** $\frac{3}{250}$
b (i) $\left(\frac{3}{250}\right)^{10}$ (ii) ${}^{10}C_5\left(\frac{3}{250}\right)^5\left(\frac{247}{250}\right)^5$ (iii) $\left(\frac{247}{250}\right)^{10} + 10\left(\frac{247}{250}\right)^9\left(\frac{3}{250}\right)$

19 **a** $\frac{7}{16}$
b (i) ${}^8C_3\left(\frac{9}{16}\right)^5\left(\frac{7}{16}\right)^3$ (ii) $1 - \left(\frac{9}{16}\right)^8 - 8\left(\frac{9}{16}\right)^7\left(\frac{7}{16}\right)^1 - {}^8C_2\left(\frac{9}{16}\right)^6\left(\frac{7}{16}\right)^2$

20 **a** 34 **b** 22

21 **a** 0.2048 **b** 0.26272 **c** (i) $\dfrac{n(n-1)(n-2)(n-3)}{20 \times 19 \times 18 \times 17}$

22 **a** $a^3 + b^3 + c^3 + 3a^2b + 3a^2c + 3ab^2 + 3ac^2 + 3b^2c + 3bc^2 + 6abc$
b (i) 0.10296 (ii) 0.13133 (iii) 0.89704

Chapter 17 Binomial Distributions - Homework Answers

Lesson 17A | Binomial probability

Past-Paper Homework - Worked Solutions

PP 1 **Northern Beaches Secondary College Manly Selective Campus, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q2**

A Bernoulli random variable comes from a Bernoulli trial, which has just two outcomes, 'success' and 'failure'. The race times in A can be any positive number, the card values in B have 13 possibilities, and the number of emails in D can be 0, 1, 2, 3, and so on. In C the result is either 'red' or 'not red', which is two outcomes. **Answer C.**

PP 2 **Blacktown Boys' High School, 2024, HSC Trial Examination, Mathematics Extension 1, Q6**

Let X be the number of hits in 10 shots, with $p = \frac{2}{3}$ and $q = \frac{1}{3}$. Then

$$P(X = 9) = {}^{10}C_9\left(\tfrac{2}{3}\right)^9\left(\tfrac{1}{3}\right)^1 = \frac{10 \times 2^9}{3^{10}} = \frac{5 \times 2^{10}}{3^{10}} = 5\left(\tfrac{2}{3}\right)^{10}.$$

Answer D. Option A has p and q the wrong way round, B leaves out ${}^{10}C_9 = 10$, and C leaves out the factor $\frac{1}{3}$ for the one miss.

PP 3 **Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q11(c)**

A success is 'a die shows a 6', so $p = \frac{1}{6}$ and $q = \frac{5}{6}$. Let X be the number of sixes on the five dice. Then

$$P(X = 2) = {}^5C_2\left(\tfrac{1}{6}\right)^2\left(\tfrac{5}{6}\right)^3 = 10 \times \frac{1}{36} \times \frac{125}{216} = \frac{625}{3888} \doteq 0.1608.$$

PP 4 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)(i)**

Each answer is correct with probability $p = \frac{1}{4}$ and incorrect with probability $q = \frac{3}{4}$, independently. Let X be the number of correct answers. Then

$$P(X = 3) = {}^5C_3\left(\tfrac{1}{4}\right)^3\left(\tfrac{3}{4}\right)^2 = 10 \times \frac{1}{64} \times \frac{9}{16} = \frac{45}{512}.$$

PP 5 **Blacktown Boys' High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q13(b)(i)(ii)**

Let X be the number of faulty cars in a box, so $n = 50$, $p = 0.03$ and $q = 0.97$.

(i) $P(X = 5) = {}^{50}C_5 \times 0.03^5 \times 0.97^{45} = 0.013\,07\ldots \doteq 0.0131$

(ii) $P(X \geq 1) = 1 - P(X = 0) = 1 - 0.97^{50} = 0.781\,93\ldots \doteq 0.7819$

PP 6 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q11(d)**

Let X be the number of races in which the pads need replacing, so $n = 12$, $p = 0.2$ and $q = 0.8$.

(i) $P(X = 3) = {}^{12}C_3 \times 0.2^3 \times 0.8^9 = 0.236\,22\ldots \doteq 0.24$

(ii) '3 occasions at most' means 0, 1, 2 or 3 occasions:

$$\begin{aligned} P(X \leq 3) &= 0.8^{12} + {}^{12}C_1 \times 0.2 \times 0.8^{11} + {}^{12}C_2 \times 0.2^2 \times 0.8^{10} + {}^{12}C_3 \times 0.2^3 \times 0.8^9 \\ &= 0.794\,56\ldots \doteq 0.79 \end{aligned}$$

PP 7 **Blacktown Boys' High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q12(a)**

Let X be the number of the 15 bags that do not contain 50 lollipops. Each bag fails with probability $p = 1 - 0.92 = 0.08$, so $q = 0.92$ and $n = 15$.

(i) $P(X = 2) = {}^{15}C_2 \times 0.08^2 \times 0.92^{13} = 0.227\,30\ldots \doteq 0.2273$

(ii) Production is stopped when $X > 2$. Use the complement:

$$\begin{aligned} P(X > 2) &= 1 - \big(P(X = 0) + P(X = 1) + P(X = 2)\big) \\ &= 1 - \big(0.92^{15} + {}^{15}C_1 \times 0.08 \times 0.92^{14} + {}^{15}C_2 \times 0.08^2 \times 0.92^{13}\big) \\ &= 0.112\,96\ldots \doteq 0.1130 \end{aligned}$$

PP 8 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q11(b)**

Two dice give 36 equally likely outcomes, and four of them total 9: $(3,6)$, $(4,5)$, $(5,4)$, $(6,3)$. So each round is a success with probability $p = \frac{4}{36} = \frac{1}{9}$, and $q = \frac{8}{9}$.

Let X be the number of successes in six rounds. Then

$$\begin{aligned} P(X \geq 5) &= P(X = 5) + P(X = 6) \\ &= {}^6C_5 \left(\tfrac{1}{9}\right)^5 \left(\tfrac{8}{9}\right)^1 + {}^6C_6 \left(\tfrac{1}{9}\right)^6 \left(\tfrac{8}{9}\right)^0 \\ &= \frac{48}{9^6} + \frac{1}{9^6} = \frac{49}{531\,441} \doteq 0.000\,092\,2. \end{aligned}$$

PP 9 **Knox Grammar School, 2022, Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)(i)(ii)**

(i) A tomato is twice as likely to come from Farm A as from Farm B, so $P(\text{A}) = \frac{2}{3}$ and $P(\text{B}) = \frac{1}{3}$. By the product rule and the addition rule,

$$\begin{aligned} P(\text{determinate}) &= P(\text{A and determinate}) + P(\text{B and determinate}) \\ &= \tfrac{2}{3} \times 0.6 + \tfrac{1}{3} \times 0.7 = \tfrac{12}{30} + \tfrac{7}{30} = \tfrac{19}{30}. \end{aligned}$$

(ii) Let X be the number of determinate tomatoes among the ten, so $p = \frac{19}{30}$ and $q = \frac{11}{30}$. The complement of 'no more than eight' is 'nine or ten', so

$$\begin{aligned} P(X \leq 8) &= 1 - P(X = 9) - P(X = 10) \\ &= 1 - {}^{10}C_9 \left(\tfrac{19}{30}\right)^9 \left(\tfrac{11}{30}\right)^1 - {}^{10}C_{10} \left(\tfrac{19}{30}\right)^{10} \left(\tfrac{11}{30}\right)^0 \\ &= 0.929\,50\ldots \doteq 0.93 \end{aligned}$$

PP 10 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q9**

Let p be the probability that a day is sunny. Three consecutive sunny days have probability $p^3 = 0.512$, so $p = \sqrt[3]{0.512} = 0.8$ and $q = 0.2$. Then

$$P(\text{exactly two sunny days in } n \text{ days}) = {}^nC_2 \times 0.8^2 \times 0.2^{n-2}.$$

Test the options: $n = 4$ gives $6 \times 0.64 \times 0.04 = 0.1536$; $n = 5$ gives $10 \times 0.64 \times 0.008 = 0.0512$; $n = 6$ gives $15 \times 0.64 \times 0.0016 = 0.015\,36$; $n = 7$ gives $21 \times 0.64 \times 0.000\,32 = 0.004\,300\,8$. Only $n = 5$ works. **Answer B.**

Chapter 17 Binomial Distributions - Homework Answers

Lesson 17B | Binomial distributions

Lesson Examples

C1 **Tutor demo.**

a (i) $X \sim B(9, \frac{1}{2})$ (ii) A 5 or a 6 has probability $\frac{2}{6} = \frac{1}{3}$, so $X \sim B(12, \frac{1}{3})$. (iii) There are 13 clubs, so $X \sim B(1, \frac{1}{4})$.

b $P(Y = 0) = 0.65$ and $P(Y = 1) = 0.35$.

$\mu = 0 \times 0.65 + 1 \times 0.35 = 0.35$

$\sigma^2 = (0^2 \times 0.65 + 1^2 \times 0.35) - 0.35^2 = 0.35 - 0.1225 = 0.2275$

$\sigma = \sqrt{0.2275} \doteq 0.477$ (These agree with $\mu = p$ and $\sigma^2 = pq = 0.35 \times 0.65$.)

Your turn.

a (i) $X \sim B(15, \frac{1}{2})$ (ii) $X \sim B(10, \frac{1}{6})$ (iii) $X \sim B(1, \frac{1}{6})$

b $P(Y = 0) = 0.2$ and $P(Y = 1) = 0.8$.

$\mu = 0 \times 0.2 + 1 \times 0.8 = 0.8$

$\sigma^2 = (0^2 \times 0.2 + 1^2 \times 0.8) - 0.8^2 = 0.8 - 0.64 = 0.16$

$\sigma = \sqrt{0.16} = 0.4$

C2 **Tutor demo.** Here $q = 1 - p = 0.8$.

a $\mu = np = 25 \times 0.2 = 5$, $\sigma^2 = npq = 25 \times 0.2 \times 0.8 = 4$, $\sigma = \sqrt{4} = 2$

b $P(X = 5) = {}^{25}C_5 \times (0.2)^5 \times (0.8)^{20} \doteq 0.196$

c

$$P(\mu - \sigma \le X \le \mu + \sigma) = P(3 \le X \le 7) = {}^{25}C_3\, p^3 q^{22} + {}^{25}C_4\, p^4 q^{21} + {}^{25}C_5\, p^5 q^{20} + {}^{25}C_6\, p^6 q^{19} + {}^{25}C_7\, p^7 q^{18} \doteq 0.793$$

d For example, a student guesses every answer in a test of 25 multiple-choice questions, each with five options, and X is the number of correct answers.

Your turn. Here $q = 1 - p = \frac{2}{3}$.

a $\mu = np = 18 \times \frac{1}{3} = 6$, $\sigma^2 = npq = 18 \times \frac{1}{3} \times \frac{2}{3} = 4$, $\sigma = \sqrt{4} = 2$

b $P(X = 6) = {}^{18}C_6 \times \left(\frac{1}{3}\right)^6 \times \left(\frac{2}{3}\right)^{12} \doteq 0.196$

c

$$P(\mu - \sigma \le X \le \mu + \sigma) = P(4 \le X \le 8) = {}^{18}C_4\, p^4 q^{14} + {}^{18}C_5\, p^5 q^{13} + {}^{18}C_6\, p^6 q^{12} + {}^{18}C_7\, p^7 q^{11} + {}^{18}C_8\, p^8 q^{10} \doteq 0.791$$

d For example, a die is thrown 18 times, and X is the number of times it shows a 5 or a 6.

C3 **Tutor demo.** Here $q = 1 - p = 0.9$.

a $\mu = np = 400 \times 0.1 = 40$, $\sigma^2 = npq = 400 \times 0.1 \times 0.9 = 36$, $\sigma = 6$

b Put $\sigma < \dfrac{n}{100}$. Squaring, $npq < \dfrac{n^2}{10\,000}$, so $n^2 > 10\,000 \times 0.09 \times n$. Hence $n > 900$, because n is positive.

c Put $\sigma < 3$. Squaring, $400p(1 - p) < 9$, so $400p^2 - 400p + 9 > 0$. The roots of $400p^2 - 400p + 9 = 0$ are $p = \dfrac{400 \pm 40\sqrt{91}}{800} = \frac{1}{2} \pm \frac{1}{20}\sqrt{91}$, that is, $p \doteq 0.0230$ and $p \doteq 0.9770$. So p should be decreased to less than $\frac{1}{2} - \frac{1}{20}\sqrt{91}$ (roughly, less than 0.023). Yes: p can be increased to more than $\frac{1}{2} + \frac{1}{20}\sqrt{91}$ (roughly, more than 0.977).

d $q = \dfrac{\sigma^2}{\mu} = \dfrac{6}{10} = 0.6$, so $p = 0.4$, and $n = \dfrac{\mu}{p} = \dfrac{10}{0.4} = 25$.

Your turn. Here $q = 1 - p = 0.8$.

a $\mu = np = 225 \times 0.2 = 45$, $\sigma^2 = npq = 225 \times 0.2 \times 0.8 = 36$, $\sigma = 6$

b Put $\sigma < \dfrac{n}{50}$. Squaring, $npq < \dfrac{n^2}{2500}$, so $n^2 > 2500 \times 0.16 \times n$. Hence $n > 400$, because n is positive.

c Put $\sigma < 3$. Squaring, $225p(1 - p) < 9$, so $25p^2 - 25p + 1 > 0$. The roots of $25p^2 - 25p + 1 = 0$ are $p = \dfrac{25 \pm 5\sqrt{21}}{50} = \frac{1}{2} \pm \frac{1}{10}\sqrt{21}$, that is, $p \doteq 0.0417$ and $p \doteq 0.9583$. So p should be decreased to less than $\frac{1}{2} - \frac{1}{10}\sqrt{21}$ (roughly, less than 0.042). Yes: p can be increased to more than $\frac{1}{2} + \frac{1}{10}\sqrt{21}$ (roughly, more than 0.958).

d $q = \dfrac{\sigma^2}{\mu} = \dfrac{3}{12} = \frac{1}{4}$, so $p = \frac{3}{4}$, and $n = \dfrac{\mu}{p} = \dfrac{12}{3/4} = 16$.

C4

Tutor demo.

a $\sigma_X^2 = 100 \times \frac{1}{2} \times \frac{1}{2} = 25$, so $\sigma_X = 5$. $\sigma_Y^2 = 100 \times \frac{1}{5} \times \frac{4}{5} = 16$, so $\sigma_Y = 4$. The ratio is $\sigma_X : \sigma_Y = 5 : 4$.

b X has $p = \frac{1}{2}$, so its distribution is symmetric about $x = 50$. Y has $p < \frac{1}{2}$, so its distribution is skewed to the right (positive skewness), with its peak near $\mu = 20$.

c $Z \sim B(100, \frac{4}{5})$, so $\mu = 100 \times \frac{4}{5} = 80$ and $\sigma = \sqrt{100 \times \frac{4}{5} \times \frac{1}{5}} = 4$. Its distribution is skewed to the left (negative skewness). Its graph is the graph for Y reflected in $x = 50$.

Your turn.

a $\sigma_X^2 = 48 \times \frac{1}{2} \times \frac{1}{2} = 12$, so $\sigma_X = 2\sqrt{3}$. $\sigma_Y^2 = 48 \times \frac{1}{4} \times \frac{3}{4} = 9$, so $\sigma_Y = 3$. The ratio is $\sigma_X : \sigma_Y = 2\sqrt{3} : 3$ (about $1.15 : 1$).

b X has $p = \frac{1}{2}$, so its distribution is symmetric about $x = 24$. Y has $p < \frac{1}{2}$, so its distribution is skewed to the right (positive skewness), with its peak near $\mu = 12$.

c $Z \sim B(48, \frac{3}{4})$, so $\mu = 48 \times \frac{3}{4} = 36$ and $\sigma = \sqrt{48 \times \frac{3}{4} \times \frac{1}{4}} = 3$. Its distribution is skewed to the left (negative skewness). Its graph is the graph for Y reflected in $x = 24$.

C5

Tutor demo.

a Here $p = q = \frac{1}{2}$ and $n = 8$, so $\mu = np = 4$, $\sigma^2 = npq = 2$, $\sigma = \sqrt{2} \doteq 1.41$.

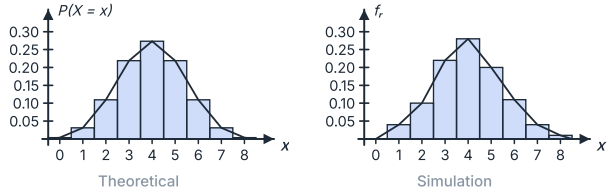
x	0	1	2	3	4	5	6	7	8
$P(X = x)$	0.004	0.031	0.109	0.219	0.273	0.219	0.109	0.031	0.004

b Each $f_r = \frac{f}{100}$:

x	0	1	2	3	4	5	6	7	8
f_r	0	0.04	0.10	0.22	0.28	0.20	0.11	0.04	0.01

$$\bar{x} = \frac{\sum xf}{\sum f} = \frac{404}{100} = 4.04 \quad \text{and} \quad s = \sqrt{\frac{\sum x^2 f}{\sum f} - \bar{x}^2} = \sqrt{\frac{1846}{100} - 4.04^2} \doteq 1.46$$

c



Your turn.

a Here $p = \frac{2}{6} = \frac{1}{3}$, $q = \frac{2}{3}$ and $n = 8$, so $\mu = np = 2.667$, $\sigma^2 = npq = 1.778$, $\sigma \doteq 1.333$.

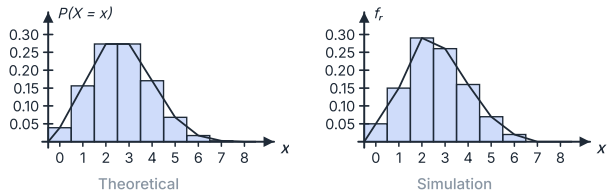
x	0	1	2	3	4	5	6	7	8
$P(X = x)$	0.039	0.156	0.273	0.273	0.171	0.068	0.017	0.002	0.000

b Each $f_r = \frac{f}{100}$:

x	0	1	2	3	4	5	6	7	8
f_r	0.05	0.15	0.29	0.26	0.16	0.07	0.02	0	0

$$\bar{x} = \frac{\sum xf}{\sum f} = \frac{262}{100} = 2.62 \quad \text{and} \quad s = \sqrt{\frac{\sum x^2 f}{\sum f} - \bar{x}^2} = \sqrt{\frac{868}{100} - 2.62^2} \doteq 1.35$$

c



Textbook Homework

1 **a** No. The events are not independent: if it rains one day, it is more likely to rain the next day, because rainy days tend to come in groups.
b Yes, and X is the number of throws that were less than five ($n = 10$, $p = \frac{2}{3}$).
c No. The possible number of trials is not fixed, and the stages are not independent, because if she wins, then the game stops.
d Yes. Let X be the number of tails turning up in 20 tosses. Then X is a random variable with a binomial distribution ($n = 20$, $p = 0.6$).
e Strictly, the pens are not replaced, so the probability changes as each pen is removed and tested. If the population of pens is large, then p is almost constant with each selection, and it could be modelled with a binomial distribution.
f No. There are not two outcomes at each stage. It could be modified to 'arrives on time' or 'takes less than 20 minutes', but the events may still not be independent.
g Yes. Note that while the experiment is different at each stage, the stages are independent and have the same probability 0.01 of success ($n = 50$, $p = 0.01$).

2 **a**

Number of heads	0	1	2	3	4	5	6	Total
Number of ways	1	6	15	20	15	6	1	64
Probability	$\frac{1}{64}$	$\frac{6}{64}$	$\frac{15}{64}$	$\frac{20}{64}$	$\frac{15}{64}$	$\frac{6}{64}$	$\frac{1}{64}$	1

b 3 heads
c $\mu = \sum xp = 3$, variance $= \sum x^2 p - \mu^2 = 10.5 - 9 = 1.5$
d $np = 6 \times \frac{1}{2} = 3$ and $npq = 6 \times \frac{1}{2} \times \frac{1}{2} = 1.5$. The results agree.
e The distribution is symmetric, thus the centre of the distribution is exactly at the midpoint.

3 **a**

x	0	1	2	3	4	5
$P(X = x)$	$\frac{1}{32}$	$\frac{5}{32}$	$\frac{10}{32}$	$\frac{10}{32}$	$\frac{5}{32}$	$\frac{1}{32}$

mode $= 2$ or 3, mean $= \frac{5}{2}$, SD $= \frac{\sqrt{5}}{2}$
b

x	0	1	2	3	4	5
$P(X = x)$	$\frac{32}{243}$	$\frac{80}{243}$	$\frac{80}{243}$	$\frac{40}{243}$	$\frac{10}{243}$	$\frac{1}{243}$

mode $= 1$ or 2, mean $= \frac{5}{3}$, SD $= \frac{\sqrt{10}}{3}$
c

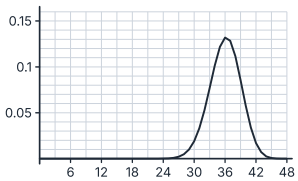
x	0	1	2	3	4	5
$P(X = x)$	0.269	0.404	0.242	0.073	0.011	0.001

mode $= 1$, mean $\doteq 1.154$, SD $\doteq 0.942$

4 **a** The expected value is 9.6 wins and the standard deviation is 2.4.
b Larry's result is 1.5 standard deviations below the mean.

5 **a** 0.263 (about 26%); $\mathrm{E}(X) = 1$, $\sigma \doteq 0.91$
b (i) 0.619 (about 62%); $\mathrm{E}(X) = 2$, $\sigma \doteq 1.29$ (ii) 0.927 (about 93%); $\mathrm{E}(X) = 4$, $\sigma \doteq 1.83$

6 **a** right skewed (positively) **b** mean 12, standard deviation 3
c The mode is 12, which has a probability of about 0.13.
d Shade the region bounded by $[6, 18]$ on the horizontal axis.
e This is the graph for $p = 0.25$ reflected horizontally in $x = 24$.

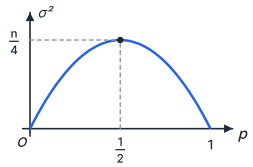


8 **a** $\mathrm{E}(X) = 12$. This is the number a student might expect to get right by guessing alone.
b 3 **c** 4 **d** $\mathrm{E}(Y) = 20$, $\sigma = 4$
e This time she is five standard deviations above the mean, which is even more unusual than her previous result.
f 75% is 36 out of 48, which is 8 standard deviations above the mean. 60% is 60 out of 100, which is 10 standard deviations above the mean. Thus it should be harder to get 60% in the second test than 75% in the first, just by chance. Note, however, that both results are almost impossible to achieve just by guessing.

9 Team A: $\mu = 35$, $\sigma \doteq 3.24$, and 45 is about 3.09σ above the mean.
Team B: $\mu = 63$, $\sigma \doteq 4.35$, and 74 is about 2.53σ above the mean.
Thus Team A's changes to the drug's delivery have shown stronger evidence for an improvement. Further trials should be carried out to check the validity of this result (see Section 17D on sampling).

10 **a** mean $= 15$, standard deviation $\doteq 3.57$
b That is, the probability of 14, 15 or 16 people voting WTP. This is about 0.325 (32.5%).

11 **a** $\sigma^2 = np(1-p) = -np^2 + np$



b It is a parabola, symmetric in its axis of symmetry $p = \frac{1}{2}$. Replacing p by $1-p$ swaps p and q, and leaves npq unchanged.
d The maximum is the vertex, occurring halfway between the roots $p = 0$ and $p = 1$, that is, at $p = \frac{1}{2}$, where $\sigma^2 = n \times \frac{1}{2} \times \frac{1}{2} = \frac{n}{4}$.
e It is clear from the quadratic graph in part **a**, and $\sigma = \sqrt{\sigma^2}$ has the same behaviour.

14 **d** If an experiment testing a certain result is repeated enough times, it is to be expected that the hypothesis will be upheld eventually. If 99 times it fails and once it succeeds, then only publishing the success gives a skewed picture of the truth.

15 **a** (iii) $\mu = \dfrac{1}{p}$
b The probability of being chosen is $p = \frac{1}{20}$, thus the mean waiting time is $\mu = \dfrac{1}{p} = 20$ time periods, or $5 \times 20 = 100$ minutes.

Chapter 17 Binomial Distributions - Homework Answers

Lesson 17B | Binomial distributions

Past-Paper Homework - Worked Solutions

PP 1 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q1**

Here $n = 12$, $p = 0.7$ and $q = 0.3$. Then

$$\mu = np = 12 \times 0.7 = 8.4 \qquad \text{and} \qquad \sigma^2 = npq = 12 \times 0.7 \times 0.3 = 2.52.$$

Answer B. The value 1.59 in A and C is about $\sqrt{2.52}$, the standard deviation, not the variance, and 12 in C and D is n, not the mean.

PP 2 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q6**

The formula is ${}^{15}C_k\, p^k q^{15-k}$ with $p = 0.29$ and $q = 0.71$, so $X \sim B(15, 0.29)$. Then

$$\mu = np = 15 \times 0.29 = 4.35.$$

Answer C. Options A and B are p and q, and $10.65 = 15 \times 0.71$ in D is the mean number of failures.

PP 3 **Blacktown Boys' High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q5**

We have $np = 225$ and $npq = 144$. Dividing the second equation by the first,

$$q = \frac{144}{225} = 0.64, \qquad \text{so} \qquad p = 1 - 0.64 = 0.36.$$

Answer D. Option B is the value of q, not p. (Also $n = 225 \div 0.36 = 625$, a whole number.)

PP 4 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(c)(i)(ii)**

Here $X \sim B(10\,000,\ 0.8)$, so $n = 10\,000$, $p = 0.8$ and $q = 0.2$.

(i) $E(X) = np = 10\,000 \times 0.8 = 8000$

(ii) $\text{Var}(X) = npq = 10\,000 \times 0.8 \times 0.2 = 1600$, so the standard deviation is $\sigma = \sqrt{1600} = 40$.

PP 5 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(c)(ii)**

Each roll is a Bernoulli trial with 'success' a sum of 7, so the number X of 7s in 30 rolls has $X \sim B(30, \frac{1}{6})$. Then

$$\text{E}(X) = np = 30 \times \tfrac{1}{6} = 5 \qquad \text{and} \qquad \text{Var}(X) = np(1-p) = 30 \times \tfrac{1}{6} \times \tfrac{5}{6} = \tfrac{25}{6}\ (\doteq 4.17).$$

PP 6 **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)(i)**

The probability that a student studies Mathematics Extension 1 is $p = 0.23$. Since $p < 0.5$, the distribution is positively skewed (skewed to the right): the peak is on the left, near the mean np, and there is a longer tail on the right-hand side.

PP 7 **Knox Grammar School, 2022, Higher School Certificate Trial Examination, Mathematics Extension 1, Q6**

The mean is $np = 0.2n$ and the standard deviation is $\sqrt{npq} = \sqrt{n \times 0.2 \times 0.8} = 0.4\sqrt{n}$. So

$$0.2n = 2 \times 0.4\sqrt{n}, \qquad 0.2n = 0.8\sqrt{n}, \qquad \sqrt{n} = 4 \ (\text{since } n \neq 0), \qquad n = 16.$$

Answer D. (Check: the mean is 3.2 and the standard deviation is $\sqrt{2.56} = 1.6$.)

PP 8 **Blacktown Boys' High School, 2022, HSC Trial Examination, Mathematics Extension 1, Q13(d)**

Here $X \sim B(30, \frac{1}{2})$, so $\mu = np = 15$, $\sigma^2 = npq = 30 \times \frac{1}{2} \times \frac{1}{2} = 7.5$ and $\sigma = \sqrt{7.5} \doteq 2.739$.

$$\begin{aligned} P(\mu - \sigma \leq X \leq \mu + \sigma) &= P(12.261 \leq X \leq 17.739) \\ &= P(X = 13,\ 14,\ 15,\ 16 \text{ or } 17) \\ &= \left({}^{30}C_{13} + {}^{30}C_{14} + {}^{30}C_{15} + {}^{30}C_{16} + {}^{30}C_{17}\right)\left(\tfrac{1}{2}\right)^{30} \\ &\doteq 0.638 \end{aligned}$$

PP 9 **Sydney Technical High School, 2023, Extension 1 Mathematics, Trial HSC, Q12(a)**

(i) $\mu = np = 12$ and $\sigma^2 = npq = 4$, where $q = 1 - p$. Dividing,

$$\frac{npq}{np} = \frac{4}{12}, \qquad q = \tfrac{1}{3}, \qquad \text{so} \qquad p = 1 - \tfrac{1}{3} = \tfrac{2}{3}.$$

(ii) $np = 12$, so $n = 12 \div \frac{2}{3} = 18$.

(iii) For example, a six-sided die is rolled 18 times and X is the number of times it shows a number greater than 2 (that is, 3, 4, 5 or 6, with probability $\frac{4}{6} = \frac{2}{3}$ on each roll).

(iv) $X \sim B(18, \frac{2}{3})$, so

$$P(X = 4) = {}^{18}C_4\left(\tfrac{2}{3}\right)^4\left(\tfrac{1}{3}\right)^{14} = \frac{3060 \times 2^4}{3^{18}} \doteq 0.000\,126.$$

PP 10 **Moriah College, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q12(a)**

(i) There are $8 + k$ marbles, and 3 of them are blue, so $P(\text{blue}) = \dfrac{3}{8+k}$.

(ii) The marble is replaced each time, so $X \sim B\left(4, \frac{3}{8+k}\right)$, with $q = 1 - \dfrac{3}{8+k} = \dfrac{5+k}{8+k}$. Then

$$\operatorname{Var}(X) = npq = 4 \times \frac{3}{8+k} \times \frac{5+k}{8+k} = \frac{12(5+k)}{(8+k)^2}.$$

Multiplying the inequality by $(8+k)^2 > 0$,

$$\begin{aligned} \frac{12(5+k)}{(8+k)^2} &< 0.8 \\ 60 + 12k &< 0.8(64 + 16k + k^2) \\ 0 &< 0.8k^2 + 0.8k - 8.8 \\ k^2 + k - 11 &> 0. \end{aligned}$$

The roots of $k^2 + k - 11 = 0$ are $k = \dfrac{-1 \pm \sqrt{45}}{2}$, that is, $k \doteq -3.85$ and $k \doteq 2.85$. Since $k \geq 0$, we need $k > 2.85$, and k is a whole number, so the smallest value is $k = 3$.

(Check: $k = 2$ gives $\operatorname{Var}(X) = \frac{12\times7}{100} = 0.84$, and $k = 3$ gives $\operatorname{Var}(X) = \frac{12\times8}{121} \doteq 0.793$.)

Chapter 17 Binomial Distributions - Homework Answers

Lesson 17C | Normal approximations to a binomial

Lesson Examples

C1 **Tutor demo.** $X \sim B\left(16, \frac{1}{2}\right)$.

a $P(X > 10) = P(X \geq 11) = \left(\frac{1}{2}\right)^{16}\left({}^{16}C_{11} + {}^{16}C_{12} + \cdots + {}^{16}C_{16}\right) = \dfrac{6885}{65\,536} \doteq 0.105$

b $\mu = np = 8$, $\sigma^2 = npq = 4$, $\sigma = 2$. With the continuity correction, $P(X \geq 11) \doteq P(X \geq 10.5)$.

$z = \dfrac{10.5 - 8}{2} = 1.25$, so $P(X \geq 10.5) \doteq P(Z \geq 1.25) = 1 - \Phi(1.25) \doteq 1 - 0.8944 = 0.1056 \doteq 0.106$.

Your turn. $X \sim B\left(12, \frac{1}{2}\right)$.

a $P(X > 8) = P(X \geq 9) = \left(\frac{1}{2}\right)^{12}\left({}^{12}C_{9} + {}^{12}C_{10} + {}^{12}C_{11} + {}^{12}C_{12}\right) = \dfrac{299}{4096} \doteq 0.073$

b $\mu = np = 6$, $\sigma^2 = npq = 3$, $\sigma = \sqrt{3} \doteq 1.732$. With the continuity correction, $P(X \geq 9) \doteq P(X \geq 8.5)$.

$z = \dfrac{8.5 - 6}{\sqrt{3}} \doteq 1.44$, so $P(X \geq 8.5) \doteq P(Z \geq 1.44) = 1 - \Phi(1.44) \doteq 1 - 0.9251 = 0.0749 \doteq 0.075$.

C2 **Tutor demo.**

a (i) $P(11.5 \leq X \leq 12.5)$ (ii) $P(X \leq 10.5)$ (iii) $P(X < 10) = P(X \leq 9) \doteq P(X \leq 9.5)$ (iv) $P(X \geq 14.5)$
(v) $P(X > 15) = P(X \geq 16) \doteq P(X \geq 15.5)$ (vi) $P(8 \leq X < 14) = P(8 \leq X \leq 13) \doteq P(7.5 \leq X \leq 13.5)$

b $\mu = 8$, $\sigma = 2$, so within one standard deviation means $6 \leq X \leq 10$.

Binomial: $P(6 \leq X \leq 10) = \left(\frac{1}{2}\right)^{16}\left({}^{16}C_{6} + {}^{16}C_{7} + {}^{16}C_{8} + {}^{16}C_{9} + {}^{16}C_{10}\right) = \dfrac{51\,766}{65\,536} \doteq 0.790$.

Normal: $P(5.5 \leq X \leq 10.5) \doteq P(-1.25 \leq Z \leq 1.25) \doteq 0.8944 - 0.1056 = 0.7888 \doteq 0.789$. The results are in excellent agreement.

Your turn.

a (i) $P(9.5 \leq X \leq 10.5)$ (ii) $P(X \leq 7.5)$ (iii) $P(X < 12) = P(X \leq 11) \doteq P(X \leq 11.5)$ (iv) $P(X \geq 12.5)$
(v) $P(X > 6) = P(X \geq 7) \doteq P(X \geq 6.5)$ (vi) $P(5 < X \leq 9) = P(6 \leq X \leq 9) \doteq P(5.5 \leq X \leq 9.5)$

b $\mu = 6$, $\sigma = \sqrt{3} \doteq 1.732$, so within one standard deviation means $4.268 \leq X \leq 7.732$, that is, $X = 5$, 6 or 7.

Binomial: $P(5 \leq X \leq 7) = \left(\frac{1}{2}\right)^{12}\left({}^{12}C_{5} + {}^{12}C_{6} + {}^{12}C_{7}\right) = \dfrac{2508}{4096} \doteq 0.612$.

Normal: $P(4.5 \leq X \leq 7.5)$: $z = \pm\dfrac{1.5}{\sqrt{3}} \doteq \pm 0.87$, so $P(-0.87 \leq Z \leq 0.87) \doteq 0.8078 - 0.1922 = 0.6156 \doteq 0.616$. The results are in good agreement.

C3 **Tutor demo.** $\mu = np = 2000$, $\sigma^2 = npq = 1600$, $\sigma = 40$.

a The z-scores are -1.25 and 1.5, so $P(1950 \leq X \leq 2060) \doteq \Phi(1.5) - \Phi(-1.25) \doteq 0.9332 - 0.1056 = 0.8276$.

b $P(1949.5 \leq X \leq 2060.5)$: the z-scores are $-1.2625 \doteq -1.26$ and $1.5125 \doteq 1.51$, so the probability is $\doteq 0.9345 - 0.1038 = 0.8307$.

The two answers differ by only 0.003, so with numbers this large the continuity correction makes very little difference.

Your turn. $\mu = np = 1200$, $\sigma^2 = npq = 900$, $\sigma = 30$.

a The z-scores are -1.67 and 2.00, so $P(1150 \leq X \leq 1260) \doteq \Phi(2.00) - \Phi(-1.67) \doteq 0.9772 - 0.0475 = 0.9297$.

b $P(1149.5 \leq X \leq 1260.5)$: the z-scores are $-1.6833 \doteq -1.68$ and $2.0167 \doteq 2.02$, so the probability is $\doteq 0.9783 - 0.0465 = 0.9318$.

The two answers differ by only 0.002, so the continuity correction is almost negligible here.

C4 **Tutor demo.** For $B\left(36, \frac{1}{2}\right)$, $\mu = 18$ and $\sigma = \sqrt{36 \times \frac{1}{2} \times \frac{1}{2}} = 3$.

${}^{36}C_{20} \times \left(\frac{1}{2}\right)^{36} = P(X = 20) \doteq P(19.5 \leq X \leq 20.5)$. The z-scores are 0.5 and $0.8333 \doteq 0.83$, so $P(19.5 \leq X \leq 20.5) \doteq \Phi(0.83) - \Phi(0.50) \doteq 0.7967 - 0.6915 = 0.1052$, and ${}^{36}C_{20} \doteq 2^{36} \times 0.1052 \doteq 7.229 \times 10^{9}$.

The calculator gives ${}^{36}C_{20} = 7\,307\,872\,110 \doteq 7.308 \times 10^{9}$, so the error is about 1%.

Your turn. For $B\left(100, \frac{1}{2}\right)$, $\mu = 50$ and $\sigma = 5$.

${}^{100}C_{55} \times \left(\frac{1}{2}\right)^{100} = P(X = 55) \doteq P(54.5 \leq X \leq 55.5)$. The z-scores are 0.9 and 1.1, so $P(54.5 \leq X \leq 55.5) \doteq \Phi(1.1) - \Phi(0.9) \doteq 0.8643 - 0.8159 = 0.0484$, and ${}^{100}C_{55} \doteq 2^{100} \times 0.0484 \doteq 6.135 \times 10^{28}$.

The calculator gives ${}^{100}C_{55} \doteq 6.145 \times 10^{28}$, so the error is about 0.2%.

C5 **Tutor demo.**

a (i) Not binomial: without replacement, the probability of a heart changes as each card is removed. (ii) Binomial, $B\left(20, \frac{1}{4}\right)$: replacing the card makes the stages identical and independent. (iii) Strictly not binomial, because the same person cannot be chosen twice, but the pool of 2 million is so large that the probabilities are virtually the same at every stage, so it is very nearly binomial.

b (i) $np = 4$, which is not greater than 5, so the approximation is not reasonable. (ii) $np = 15$ and $nq = 45$, both greater than 5, so it is reasonable. (iii) $np = 196$ but $nq = 4$, so it is not reasonable.

c $np = 0.04n > 5$ gives $n > 125$ (and then $nq = 0.96n > 5$ also holds), so $n > 125$.

Your turn.

a (i) Not binomial: without replacement, the probability of a red marble changes at each draw, and the bag is small. (ii) Binomial, $B\left(30, \frac{1}{6}\right)$. (iii) Strictly not binomial, but 40 is small relative to 1500, so it is very nearly binomial.

b (i) $np = 7.5$ and $nq = 17.5$, so it is reasonable. (ii) $np = 3$, so it is not reasonable. (iii) $np = 27$ but $nq = 3$, so it is not reasonable.

c $nq = 0.05n > 5$ gives $n > 100$ (and then $np = 0.95n > 5$ also holds), so $n > 100$.

Textbook Homework

Normal probabilities below use the standard normal table with z correct to two decimal places. Answers may vary slightly with interpolation or technology.

1 **a** $X \sim B(20, 0.3)$ **b** $P(9 \le X \le 11) = {}^{20}C_9(0.3)^9(0.7)^{11} + {}^{20}C_{10}(0.3)^{10}(0.7)^{10} + {}^{20}C_{11}(0.3)^{11}(0.7)^9 \doteq 10.82\%$

c $np = 6 > 5$ and $nq = 14 > 5$.

d $\mu = 6$, $\sigma^2 = 4.2$, $\sigma \doteq 2.049$. The z-scores are $\dfrac{8.5 - 6}{\sqrt{4.2}} \doteq 1.22$ and $\dfrac{11.5 - 6}{\sqrt{4.2}} \doteq 2.68$, so $P(8.5 \le X \le 11.5) \doteq 0.9963 - 0.8888 = 0.1075 = 10.75\%$.

e $\dfrac{10.82 - 10.75}{10.82} \doteq 0.65\%$. The result is very accurate.

2 **a** $B(50, 0.5)$, $P(18 \le X \le 20) \doteq 8.49\%$; $np = 25$, $nq = 25$; normal approximation using $N(25, 12.5)$: $z = -2.12$ and -1.27, $0.1020 - 0.0170 = 8.50\%$, percentage error 0.1%

b $B(20, 0.4)$, $P(8 \le X \le 9) \doteq 33.94\%$; $np = 8$, $nq = 12$; normal approximation using $N(8, 4.8)$: $z = -0.23$ and 0.68, $0.7517 - 0.4090 = 34.27\%$, percentage error 1.0%

c $B(30, 0.3)$, $P(5 \le X \le 7) \doteq 25.12\%$; $np = 9$, $nq = 21$; normal approximation using $N(9, 6.3)$: $z = -1.79$ and -0.60, $0.2743 - 0.0367 = 23.76\%$, percentage error 5.4%

d $B(40, 0.2)$, $P(9 \le X \le 12) \doteq 36.36\%$; $np = 8$, $nq = 32$; normal approximation using $N(8, 6.4)$: $z = 0.20$ and 1.78, $0.9625 - 0.5793 = 38.32\%$, percentage error 5.4%

e $B(22, 0.6)$, $P(13 \le X \le 15) \doteq 46.59\%$; $np = 13.2$, $nq = 8.8$; normal approximation using $N(13.2, 5.28)$: $z = -0.30$ and 1.00, $0.8413 - 0.3821 = 45.92\%$, percentage error 1.4%

f $B(80, 0.1)$, $P(10 \le X \le 13) \doteq 24.98\%$; $np = 8$, $nq = 72$; normal approximation using $N(8, 7.2)$: $z = 0.56$ and 2.05, $0.9798 - 0.7123 = 26.75\%$, percentage error 7.1%

g $B(500, 0.25)$, $P(100 \le X \le 103) \doteq 0.84\%$; $np = 125$, $nq = 375$; normal approximation using $N(125, 93.75)$: $z = -2.63$ and -2.22, $0.0132 - 0.0043 = 0.89\%$, percentage error 6.0%

h $B(200, 0.9)$, $P(170 \le X \le 172) \doteq 3.39\%$; $np = 180$, $nq = 20$; normal approximation using $N(180, 18)$: $z = -2.47$ and -1.77, $0.0384 - 0.0068 = 3.16\%$, percentage error 6.8%

3 **a** There are two possible outcomes, pink or blue. **b** There are 20 stages, each independent, and with the same probability of success.

c If the counter is not returned, the stages of the experiment will not be independent. With the large number of counters, however, the probability will not change much, and we could approximate the experiment as binomial.

d $p = 0.6$, $\mu = np = 12$, $\sigma^2 = np(1-p) = 4.8$, so $\sigma \doteq 2.19$. **e** ${}^{20}C_{14}(0.6)^{14}(0.4)^6 \doteq 0.124$

f (i) The column for 14 pink counters is 1 unit wide and centred on 14, so it runs from 13.5 to 14.5, and the normal curve covers the same interval. (ii) $P(13.5 < X < 14.5) \doteq P(0.68 < Z < 1.14) \doteq 0.8729 - 0.7517 = 0.1212 \doteq 0.121$. Correct to two significant figures, both are 0.12, so they are in strong agreement.

5 **a** $n > 10$ **b** $n > 20$ **c** $n > 40$ **d** $n > 500$ **e** $n > 20$ **f** $n > 40$ **g** $n > 50$ **h** $n > 11.1$, that is, $n \ge 12$

6 **a** Very large numbers are involved. A normal calculator cannot handle numbers such as ${}^{854}C_{76}$, and other calculating devices may not be accurate with such large numbers. There are also a large number of cases to consider, from 60 to 76 successes.
b It is hard to get a representative sample of the whole world, because different ethnic groups will have different tendencies to colour-blindness.
c $\mu = 854 \times 0.08 = 68.32$, $\sigma = \sqrt{854 \times 0.08 \times 0.92} \doteq 7.93$. Here 7% of 854 is 59.78 and 9% is 76.86, so we need 60 to 76 people.
$P(60 < X < 76) \doteq P(-1.05 < Z < 0.97) \doteq 0.8340 - 0.1469 = 0.6871 \doteq 68.7\%$
d $P(76 < X < 76.5) \doteq P(0.97 < Z < 1.03) \doteq 0.8485 - 0.8340 = 0.0145 \doteq 1.5\%$

7 **a** $\left(\frac{3}{4}\right)^{15} \doteq 0.0134 \doteq 1\%$ **b** $1 - \left(\frac{3}{4}\right)^{15} \doteq 99\%$
c Twenty per cent of 15 is 3. $\mu = 3.75$, $\sigma^2 = 2.8125$, $\sigma \doteq 1.677$.
$P(X \geq 3) \doteq P(X \geq 2.5) \doteq P(Z \geq -0.75) \doteq 0.7734 \doteq 77\%$

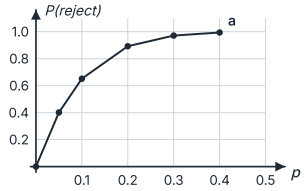
8 Interpreting this as $P(11 \leq X \leq 20)$, where $X \sim B(20, 0.45)$: $\mu = 9$, $\sigma \doteq 2.225$, and $P(X \geq 10.5) \doteq P(Z \geq 0.67) \doteq 1 - 0.7486 = 0.2514$, so a normal approximation gives 25%.

9 $X \sim B(30, 0.6)$: $\mu = 18$, $\sigma \doteq 2.683$.
$P(X > 20) = P(21 \leq X \leq 30) \doteq P(X \geq 20.5) \doteq P(Z \geq 0.93) \doteq 1 - 0.8238 = 0.1762 \doteq 17.6\%$.
The underlying Bernoulli distribution is not applied with replacement, because the same person will not be in the park twice at the same gathering. If the population of Nashville is large, it should be reasonable to neglect this fact. It is also assumed that the visitors to the park are a random cross-section of Nashville. Groups with similar musical tastes may arrive together.

12 **a** (i) $1 - 0.95^{10} \doteq 0.40$
(ii) $P(\text{reject}) = 1 - (1-p)^{10}$:

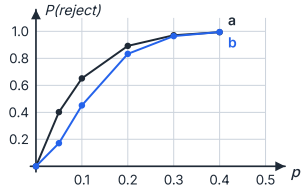
p	0	0.05	0.1	0.2	0.3	0.4
$P(\text{reject})$	0	0.40	0.65	0.89	0.97	0.99

(iii)



b $P(\text{reject}) = 1 - (1-p)^{15} - 15p(1-p)^{14}$:

p	0	0.05	0.1	0.2	0.3	0.4
$P(\text{reject})$	0	0.17	0.45	0.83	0.96	0.99



c The second method is more forgiving if there are a few punnets that need to be rejected. Both methods are strongly likely to reject the batch if p is high, indeed the curves approach one another closely by the time p reaches 20%.

Chapter 17 Binomial Distributions - Homework Answers

Lesson 17C | Normal approximations to a binomial

Past-Paper Homework - Worked Solutions

PP 1 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(c)(iii)**

Using the normal approximation without a continuity correction, as in the school solution.

X is approximately $N(8000, 40^2)$, so

$$P(X < 8040) \doteq P\left(Z < \frac{8040 - 8000}{40}\right) = P(Z < 1) \doteq 0.8413 \doteq 0.84.$$

(The empirical rule gives the same answer: $0.5 + 0.34 = 0.84$.)

PP 2 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q12(b)**

Using the normal approximation without a continuity correction, as in the school solution.

Let X be the number of patients who cannot be cured. Then $X \sim B(150, 0.2)$, with $np = 30$ and $nq = 120$. Both are greater than 5 (indeed greater than 10), so the normal approximation is justified.

$$\mu = np = 30, \qquad \sigma = \sqrt{npq} = \sqrt{150 \times 0.2 \times 0.8} = \sqrt{24} \doteq 4.899$$

$$P(X \leq 37) \doteq P\left(Z \leq \frac{37 - 30}{\sqrt{24}}\right) \doteq P(Z \leq 1.43) \doteq 0.9236.$$

(The school solution works with the number cured, $P(X \geq 113)$, and gets the same z-score, -1.43.)

PP 3 **Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q11(e)**

Using the normal approximation to the count, without a continuity correction. (The school solution's main method uses the sample proportion, Lesson 17D, and gets the same z-score. Its alternatives include the count without a continuity correction, as here, and the count with one, which gives 0.0336.)

Let X be the number of left-handed people. Then $X \sim B(100, 0.1)$, with $np = 10$ and $nq = 90$, both greater than 5.

$$\mu = np = 10, \qquad \sigma^2 = npq = 9, \qquad \sigma = 3.$$

5% of one hundred is 5, so we need $P(X < 5)$:

$$\begin{aligned} P(X < 5) &\doteq P\left(Z < \frac{5 - 10}{3}\right) \\ &= P(Z < -1.67) \\ &= 1 - \Phi(1.67) \\ &\doteq 1 - 0.9525 \\ &= 0.0475. \end{aligned}$$

PP 4 **Killara High School, 2020, Mathematics Extension 1 Trial HSC Examination, Q12(e)**

Using the normal approximation without a continuity correction, as in the school solution.

Let X be the number who order dessert. Then $X \sim B(288, 0.27)$, with $np = 77.76$ and $nq = 210.24$, both greater than 5.

$$\mu = np = 77.76, \qquad \sigma^2 = npq = 56.7648, \qquad \sigma \doteq 7.534$$

$$P(X < 100) \doteq P\left(Z < \frac{100 - 77.76}{7.534}\right) \doteq P(Z < 2.95) \doteq 0.9984 \doteq 0.998.$$

PP 5 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q14(c)(ii)**

Using the normal approximation to the count, without a continuity correction. (The school solution uses the sample proportion, Lesson 17D, and gets the same z-score.)

The 2 million trials estimate $p \doteq \dfrac{1\,615\,580}{2\,000\,000} = 0.80779$. In 10 000 trials, $X \sim B(10\,000,\ 0.80779)$, so

$$\mu = np = 8077.9, \qquad \sigma = \sqrt{10\,000 \times 0.80779 \times 0.19221} \doteq 39.40.$$

$$\begin{aligned} P(X > 8100) &\doteq P\left(Z > \frac{8100 - 8077.9}{39.40}\right) \\ &\doteq P(Z > 0.56) \\ &= 1 - 0.7123 \\ &= 0.2877 \doteq 28.8\%. \end{aligned}$$

PP 6 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q13(d)**

Using the normal approximation without a continuity correction, as in the school solution's main method.

Let N be the number absent, so $N \sim B(n, 0.05)$, with $\mu = 0.05n$ and $\sigma = \sqrt{0.05 \times 0.95\, n} = \sqrt{0.0475n}$.

From the table, $\Phi(1.32) \doteq 0.9066$, so $P(N \leq 20) = 0.90658$ gives $z = 1.32$:

$$\frac{20 - 0.05n}{\sqrt{0.0475n}} = 1.32, \qquad \text{so} \qquad 0.05n + 1.32\sqrt{0.0475}\,\sqrt{n} - 20 = 0.$$

This is a quadratic in $u = \sqrt{n}$: $0.05u^2 + 0.28769u - 20 = 0$. Since $u > 0$,

$$u = \frac{-0.28769 + \sqrt{0.28769^2 + 4}}{0.1} \doteq 17.329, \qquad n = u^2 \doteq 300.3, \qquad \text{so} \qquad n = 300.$$

(Check: $np = 15$ and $nq = 285$, so the normal approximation is reasonable. The school solution also gives the continuity-correction version, with 20.5 in place of 20, which leads to $n \doteq 309$.)

PP 7 **James Ruse Agricultural High School, 2023, Year 12 Trial HSC Examination, Mathematics Extension 1, Q13(b)(ii)**

Using the normal approximation without a continuity correction, as in the school solution.

Let X be the number who take less than 12 minutes, so $X \sim B(n, 0.1)$, with $\mu = 0.1n$ and $\sigma = \sqrt{0.1 \times 0.9\, n} = 0.3\sqrt{n}$.

From the table, $\Phi(0.40) \doteq 0.6554$, so $\Phi(-0.40) \doteq 0.3446$. Hence $P(X < 9) = 0.3446$ gives $z = -0.4$:

$$\frac{9 - 0.1n}{0.3\sqrt{n}} = -0.4, \qquad \text{so} \qquad 0.1n - 0.12\sqrt{n} - 9 = 0.$$

This is a quadratic in $u = \sqrt{n}$: $0.1u^2 - 0.12u - 9 = 0$. Since $u > 0$,

$$u = \frac{0.12 + \sqrt{0.0144 + 3.6}}{0.2} \doteq 10.106, \qquad n = u^2 \doteq 102.13, \qquad \text{so} \qquad n = 102.$$

(The school solution also squares the equation, giving $0.01n^2 - 1.8144n + 81 = 0$, with roots $n \doteq 102.13$ and $n \doteq 79.31$. It rejects $n = 79$, because then $\mu = 7.9$ is below 9, so the z-score of 9 would be positive, not -0.4.)

Chapter 17 Binomial Distributions - Homework Answers

Lesson 17D | Sample proportions

Lesson Examples

C1 **Tutor demo.** $X \sim B\left(3, \frac{1}{3}\right)$, and $P(X = x) = {}^3C_x\left(\frac{1}{3}\right)^x\left(\frac{2}{3}\right)^{3-x}$.

a

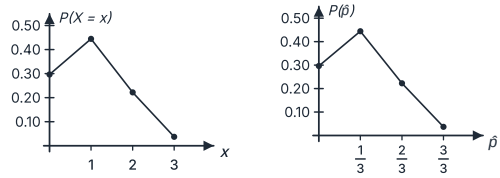
x	0	1	2	3
prob	$\frac{8}{27}$	$\frac{12}{27}$	$\frac{6}{27}$	$\frac{1}{27}$

$E(X) = np = 1$ and $\mathrm{Var}(X) = npq = \frac{2}{3}$.

b $\hat{p} = \dfrac{X}{3}$, with the same probabilities:

$\hat{p}$	0	$\frac{1}{3}$	$\frac{2}{3}$	1
prob	$\frac{8}{27}$	$\frac{12}{27}$	$\frac{6}{27}$	$\frac{1}{27}$

$E(\hat{p}) = p = \frac{1}{3}$ and $\mathrm{Var}(\hat{p}) = \dfrac{pq}{n} = \frac{2}{27}$.



Your turn. $X \sim B\left(4, \frac{1}{2}\right)$, and $P(X = x) = {}^4C_x\left(\frac{1}{2}\right)^4$.

a

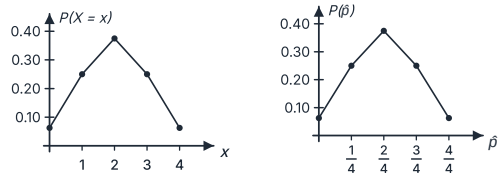
x	0	1	2	3	4
prob	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

$E(X) = np = 2$ and $\mathrm{Var}(X) = npq = 1$.

b $\hat{p} = \dfrac{X}{4}$, with the same probabilities:

$\hat{p}$	0	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	1
prob	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

$E(\hat{p}) = p = \frac{1}{2}$ and $\mathrm{Var}(\hat{p}) = \dfrac{pq}{n} = \frac{1}{16}$.



C2 **Tutor demo.** $X \sim B\left(12, \frac{1}{4}\right)$ and $\hat{p} = \dfrac{X}{12}$.

a Within 0.05 of 0.25 means $0.2 \le \hat{p} \le 0.3$. Only $\hat{p} = \frac{3}{12} = 0.25$ lies there ($\frac{2}{12} \doteq 0.167$ and $\frac{4}{12} \doteq 0.333$ do not).

$P\left(\hat{p} = \frac{3}{12}\right) = {}^{12}C_3\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^9 \doteq 0.258$

b Within 0.1 of 0.25 means $0.15 \le \hat{p} \le 0.35$, so $\hat{p} = \frac{2}{12}$, $\frac{3}{12}$ or $\frac{4}{12}$.

$P = {}^{12}C_2\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^{10} + {}^{12}C_3\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^9 + {}^{12}C_4\left(\frac{1}{4}\right)^4\left(\frac{3}{4}\right)^8 \doteq 0.2323 + 0.2581 + 0.1936 \doteq 0.684$

Your turn. $X \sim B\left(15, \frac{2}{3}\right)$ and $\hat{p} = \dfrac{X}{15}$.

a Within 0.05 of $\frac{2}{3} \doteq 0.667$: only $\hat{p} = \frac{10}{15}$ lies there ($\frac{9}{15} = 0.6$ and $\frac{11}{15} \doteq 0.733$ do not).

$P\left(\hat{p} = \frac{10}{15}\right) = {}^{15}C_{10}\left(\frac{2}{3}\right)^{10}\left(\frac{1}{3}\right)^5 \doteq 0.214$

b Within 0.1 of $\frac{2}{3}$: $\hat{p} = \frac{9}{15}$, $\frac{10}{15}$ or $\frac{11}{15}$ ($\frac{8}{15} \doteq 0.533$ and $\frac{12}{15} = 0.8$ are too far away).

$P = {}^{15}C_9\left(\frac{2}{3}\right)^9\left(\frac{1}{3}\right)^6 + {}^{15}C_{10}\left(\frac{2}{3}\right)^{10}\left(\frac{1}{3}\right)^5 + {}^{15}C_{11}\left(\frac{2}{3}\right)^{11}\left(\frac{1}{3}\right)^4 \doteq 0.1786 + 0.2143 + 0.1948 \doteq 0.588$

C3 **Tutor demo.** $\hat{p}$ is approximately $N\left(0.4, \dfrac{0.4 \times 0.6}{1000}\right)$, so $\sigma = \sqrt{0.00024} \doteq 0.01549$.

a $z = \pm 0.02 \div 0.01549 \doteq \pm 1.29$, so $P(0.38 \leq \hat{p} \leq 0.42) \doteq 0.9015 - 0.0985 = 0.8030$.

b $z = \pm 0.03 \div 0.01549 \doteq \pm 1.94$, so $P(0.37 \leq \hat{p} \leq 0.43) \doteq 0.9738 - 0.0262 = 0.9476$.

c $P(\hat{p} > 0.42) \doteq P(Z > 1.29) = 1 - 0.9015 = 0.0985$.

Your turn. $\hat{p}$ is approximately $N\left(0.3, \dfrac{0.3 \times 0.7}{1500}\right)$, so $\sigma = \sqrt{0.00014} \doteq 0.01183$.

a $z = \pm 0.01 \div 0.01183 \doteq \pm 0.85$, so $P(0.29 \leq \hat{p} \leq 0.31) \doteq 0.8023 - 0.1977 = 0.6046$.

b $z = \pm 0.02 \div 0.01183 \doteq \pm 1.69$, so $P(0.28 \leq \hat{p} \leq 0.32) \doteq 0.9545 - 0.0455 = 0.9090$.

c $P(\hat{p} < 0.28) \doteq P(Z < -1.69) = 1 - 0.9545 = 0.0455$.

C4 **Tutor demo.** Divide the values of x by 8.

a

$\hat{p}$	0	0.125	0.25	0.375	0.5	0.625	0.75	0.875	1
f_r	0	0.04	0.10	0.22	0.28	0.20	0.11	0.04	0.01
cf_r	0	0.04	0.14	0.36	0.64	0.84	0.95	0.99	1.00

From the data, $\bar{x} = \dfrac{4.04}{8} = 0.505$ and $s \doteq \dfrac{1.462}{8} \doteq 0.183$.

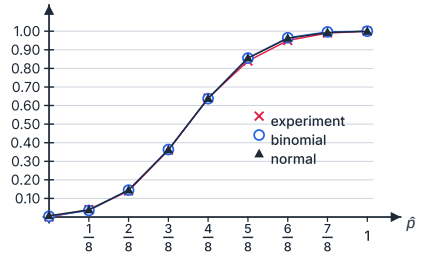
b $\mu = p = 0.5$ and $\sigma = \sqrt{\dfrac{0.5 \times 0.5}{8}} \doteq 0.177$.

$\hat{p}$	0	0.125	0.25	0.375	0.5	0.625	0.75	0.875	1
Cmlve	0.004	0.035	0.145	0.363	0.637	0.855	0.965	0.996	1.000

c The same mean 0.5 and standard deviation 0.177. The gaps between the values of $\hat{p}$ are $\frac{1}{8}$, so for each value we find the z-score of $\hat{p} + \frac{1}{16} = \hat{p} + 0.0625$.

$\hat{p}$	0	0.125	0.25	0.375	0.5	0.625	0.75	0.875	1
z	−2.475	−1.768	−1.061	−0.354	0.354	1.061	1.768	2.475	3.182
$\Phi(z)$	0.007	0.039	0.144	0.362	0.638	0.856	0.961	0.993	0.999

d



Your turn. Divide the values of x by 8.

a

$\hat{p}$	0	0.125	0.25	0.375	0.5	0.625	0.75	0.875	1
f_r	0.05	0.15	0.29	0.26	0.16	0.07	0.02	0	0
cf_r	0.05	0.20	0.49	0.75	0.91	0.98	1.00	1.00	1.00

From the data, $\bar{x} = \dfrac{2.62}{8} = 0.3275$ and $s \doteq \dfrac{1.347}{8} \doteq 0.168$.

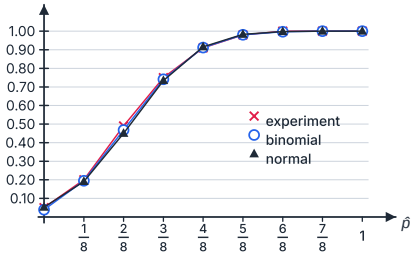
b $\mu = p = \frac{1}{3} \doteq 0.333$ and $\sigma = \sqrt{\dfrac{\frac{1}{3} \times \frac{2}{3}}{8}} = \frac{1}{6} \doteq 0.167$.

$\hat{p}$	0	0.125	0.25	0.375	0.5	0.625	0.75	0.875	1
Cmlve	0.039	0.195	0.468	0.741	0.912	0.980	0.997	1.000	1.000

c The same mean $\frac{1}{3}$ and standard deviation $\frac{1}{6}$. For each value we find the z-score of $\hat{p} + 0.0625$.

$\hat{p}$	0	0.125	0.25	0.375	0.5	0.625	0.75	0.875	1
z	−1.625	−0.875	−0.125	0.625	1.375	2.125	2.875	3.625	4.375
$\Phi(z)$	0.052	0.191	0.450	0.734	0.915	0.983	0.998	1.000	1.000

d



Textbook Homework

Normal probabilities below use the standard normal table with z correct to two decimal places. Answers may vary slightly with interpolation or technology.

1 **a** $\frac{2}{5}$ **b** $\frac{4}{10} = \frac{2}{5}$ **c** $\frac{9}{12} = \frac{3}{4}$

2 **a**

x	0	1	2	3	4	5
$P(X = x)$	$\frac{1}{32}$	$\frac{5}{32}$	$\frac{10}{32}$	$\frac{10}{32}$	$\frac{5}{32}$	$\frac{1}{32}$

b

$\hat{p}$	0	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	1
$P(\hat{p})$	$\frac{1}{32}$	$\frac{5}{32}$	$\frac{10}{32}$	$\frac{10}{32}$	$\frac{5}{32}$	$\frac{1}{32}$

c 0.5 **d** It is the probability p in each Bernoulli trial, that is, it is the probability of a coin landing heads.

3 **a**

$\hat{p}$	0	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	1
f_r	$\frac{1}{20}$	$\frac{1}{20}$	$\frac{3}{20}$	$\frac{2}{20}$	$\frac{6}{20}$	$\frac{7}{20}$

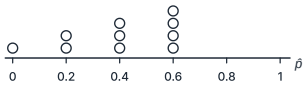
b 0.72 **c** It is an estimate of the probability that a shopper chosen at random lives in the suburb.

4 **a** 8, 9 or 10 heads **b** $P(X \geq 8) = \dfrac{{}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10}}{2^{10}} = \dfrac{56}{1024} \doteq 5.47\%$

6 **a** $12 \div 32 = 0.375$ **b** 0.2 (only Dakarai buys lunch regularly)
c The ten samples give $\hat{p} = 0.2,\ 0.4,\ 0.2,\ 0.6,\ 0.6,\ 0.4,\ 0.4,\ 0.6,\ 0,\ 0.6$.

$\hat{p}$	0	0.2	0.4	0.6	0.8	1.0
Frequency	1	2	3	4	0	0

d



7 $\mu = 0.2$, $\sigma = \sqrt{0.2 \times 0.8 \div 500} \doteq 0.01789$. $P(\hat{p} > 0.22) \doteq P(Z > 1.12) = 1 - 0.8686 = 0.1314 \doteq 13\%$

8 **a** Let X record the number of hearts selected, $X \sim B(80, 0.25)$, $\mu = 20$, $\sigma = \sqrt{15} \doteq 3.873$. Then
$P(15.5 \leq X \leq 24.5) \doteq P(-1.16 \leq Z \leq 1.16) = 2 \times 0.8770 - 1 = 0.7540 \doteq 75\%$.
b Let $\mu = 0.25$, and $\sigma = \sqrt{0.25 \times 0.75 \div 80} \doteq 0.0484$. Thus
$P(0.20 \leq \hat{p} \leq 0.30) \doteq P(-1.03 \leq Z \leq 1.03) = 2 \times 0.8485 - 1 = 0.6970 \doteq 70\%$. (The exact answer is 76%, so part **a** is more accurate.)

9 $\mu = 0.7$, $\sigma = \sqrt{0.7 \times 0.3 \div 300} \doteq 0.02646$.
a $P(\hat{p} \geq 0.65) \doteq P(Z \geq -1.89) = 0.9706 \doteq 97\%$ **b**
$P(0.65 \leq \hat{p} \leq 0.75) \doteq P(-1.89 \leq Z \leq 1.89) = 2 \times 0.9706 - 1 = 0.9412 \doteq 94\%$

10 **a** (i) $0.95^{100} \doteq 0.0059 \doteq 0.6\%$ (ii) Fewer than 2 people react: $0.95^{100} + 100 \times 0.05 \times 0.95^{99} \doteq 0.0371 \doteq 3.7\%$
b (i) The mean and standard deviation for $\hat{p}$ are $\mu = 0.05$, and $\sigma = \sqrt{0.05 \times 0.95 \div 1000} \doteq 0.0069$. Thus
$P(\hat{p} < 0.03) \doteq P(Z < -2.90) = 1 - 0.9981 = 0.0019 \doteq 0.2\%$.
(ii) This result is significantly different from the previous claim that 5% of patients will have a reaction. They should check whether the sample was random: perhaps it consisted of patients more resistant to the side effects of the medication. They should also check whether there have been any changes to the medication to reduce patient reactions. It is also possibly just chance that this result occurred, but the likelihood of this is small.

11 **a** $\mu = 0.6$, $\sigma^2 = 0.012$, so $\sigma \doteq 0.1095$.

b $\hat{p}$ can take only the 21 separate values $0,\ \frac{1}{20},\ \frac{2}{20},\ \ldots,\ 1$, and no values in between, so its distribution is discrete.

c $P(\hat{p} \le 0.4) \doteq P(Z \le -1.83) = 1 - 0.9664 = 0.0336 \doteq 3.4\%$

d $\dfrac{5.7 - 3.4}{5.7} \doteq 40\%$. No, there is an error of about 40%. The sample is too small and we are not using any continuity correction.

e $P(\hat{p} \le 0.75) \doteq P(Z \le 1.37) = 0.9147 \doteq 91.5\%$

f $\dfrac{95 - 91.5}{95} \doteq 3.7\%$. The curve is flatter at the top end and varies less with $\hat{p}$. The percentage difference is also exaggerated by small values, such as at the left end of the curve.

12 **a, b** $\sigma = \dfrac{0.5}{\sqrt{n}}$ and $z = \dfrac{0.52 - 0.5}{\sigma}$ ($z \doteq 1.26$, 0.89, 0.40, 0.28, 0.20).

n	1000	500	100	50	25
exact	0.9026	0.8262	0.6914	0.6641	0.6550
approx	0.8962	0.8133	0.6554	0.6103	0.5793
% error	0.7	1.6	5.2	8.1	11.6

c The accuracy improves as n increases, and is quite good for large samples.

13 For the sample proportion $\hat{p}$, $\mu = 0.5$, and $\sigma = \sqrt{0.5 \times 0.5 \div 50} \doteq 0.0707$. Thus $P(\hat{p} > 0.6) \doteq P(Z > 1.41) = 1 - 0.9207 = 0.0793 \doteq 8\%$. It appears that people strongly prefer the branded version, even though it is identical. There may be an expectation that the branded version is superior, or they may prefer the packaging.

14 In the population of those with the disease, let $p = 0.3$ be the probability of a positive response to the drug. Let X be the binomial random variable measuring how many respond positively to the drug in the trial of $n = 100$, and let $\hat{p} = \dfrac{X}{n}$ be the sample proportion. The mean and standard deviation of $\hat{p}$ are $\mu = p = 0.3$ and $\sigma = \sqrt{p(1-p)/n} \doteq 0.0458$. Then $P(\hat{p} \ge 0.4) \doteq P(Z \ge 2.18) = 1 - 0.9854 = 0.0146 \doteq 1.5\%$. It is unlikely to be by chance.

15 We need $\dfrac{pq}{n} < \left(\dfrac{k}{100}\right)^2$, that is, $n > \dfrac{10\,000\,pq}{k^2}$.

a (i) $n > 131.25$, so 132 (ii) $n > 233.3$, so 234 (iii) $n > 525$, so 526 (iv) $n > 2100$, so 2101 (v) $n > \dfrac{2100}{k^2}$

b (i) $n > 100$, so 101 (ii) $n > 177.8$, so 178 (iii) $n > 400$, so 401 (iv) $n > 1600$, so 1601 (v) $n > \dfrac{1600}{k^2}$

c (i) $n > 156.25$, so 157 (ii) $n > 277.8$, so 278 (iii) $n > 625$, so 626 (iv) $n > 2500$, so 2501 (v) $n > \dfrac{2500}{k^2}$

16 **a** (i) $\dfrac{{}^{10}C_6}{2^{10}} = \dfrac{210}{1024} \doteq 0.205$ (ii) $40 \times 0.205 \doteq 8$

17 **a** $p = \frac{6}{36} = \frac{1}{6}$ (the sums 10, 11 and 12 occur in 3, 2 and 1 ways)

b $X \sim B(20, \frac{1}{6})$: $P(0 \le X \le 4) \doteq 0.7687$

c $\mu = \frac{10}{3}$, $\sigma = \sqrt{20 \times \frac{1}{6} \times \frac{5}{6}} = \frac{5}{3}$. $P(X \le 4) \doteq P(Z \le 0.40) = 0.6554$

d $P(X \le 4.5) \doteq P(Z \le 0.70) = 0.7580$

e $\mu = \frac{1}{6}$, $\sigma = \sqrt{\frac{1}{6} \times \frac{5}{6} \div 20} = \frac{1}{12}$. $P(\hat{p} \le 0.2) \doteq P(Z \le 0.40) = 0.6554$. This agrees with part **c**: since $\hat{p} = \dfrac{X}{20}$, the event $\hat{p} \le 0.2$ is the event $X \le 4$, and the sample proportion distribution is just the binomial compressed horizontally by a factor $\dfrac{1}{n}$. Its mean and standard deviation are both divided by 20, so the z-score, and the corresponding area, are the same.

18 **b** (i) $\sigma = \sqrt{0.67 \times 0.33 \div 100} \doteq 0.047$

(ii) Margin of error $= 1.96\sigma \doteq 0.09 = 9\%$, and the 95% confidence interval for the probability p is $[58\%,\ 76\%]$.

(iii) $1.96\sqrt{\dfrac{0.67 \times 0.33}{n}} \le 0.01$ gives $n \ge 8493.8$, so $n \ge 8494$.

Chapter 17 Binomial Distributions - Homework Answers

Lesson 17D | Sample proportions

Past-Paper Homework - Worked Solutions

PP 1 **Baulkham Hills High School, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q4**

Answer B. The standard deviation of $\hat{p}$ is $\sqrt{\dfrac{pq}{n}}$, so we need

$$\sqrt{\frac{0.3\times 0.7}{n}} \le 0.07, \qquad \text{so} \qquad \frac{0.21}{n} \le 0.0049, \qquad \text{so} \qquad n \ge \frac{0.21}{0.0049} \doteq 42.86.$$

The smallest value is $n = 43$. (With $n = 42$, the standard deviation is about 0.0707, which is more than 0.07, so A is wrong.)

PP 2 **Killara High School, 2021, Year 12 Mathematics Extension 1 Trial Examination, Q12(a)(ii)**

Here $n = 150$ and $p = 0.9$.

$$E(\hat{p}) = p = 0.9, \qquad \operatorname{Var}(\hat{p}) = \frac{0.9\times(1-0.9)}{150} = \frac{3}{5000}, \qquad \text{SD} = \sqrt{\frac{3}{5000}} \doteq 0.0245.$$

PP 3 **James Ruse Agricultural High School, 2023, Year 12 Trial HSC Examination, Mathematics Extension 1, Q15(c)**

Using the normal approximation to $\hat{p}$ without a continuity correction, as in the school solution.

(i) $np = 100\times 0.2 = 20$ and $nq = 100\times 0.8 = 80$. Both are greater than 5 (indeed at least 10), so the proportion of heads can be approximated using the normal distribution.

(ii) $\mu = p = 0.2$ and $\sigma = \sqrt{\dfrac{0.2\times 0.8}{100}} = 0.04$.

(iii)

$$\begin{aligned} P(0.1 \le \hat{p} \le 0.3) &\doteq P\left(\frac{0.1-0.2}{0.04} \le Z \le \frac{0.3-0.2}{0.04}\right) \\ &= P(-2.5 \le Z \le 2.5) \\ &= 0.9938 - 0.0062 \\ &= 0.9876. \end{aligned}$$

PP 4 **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)(ii)**

Using the normal approximation to $\hat{p}$ without a continuity correction, as in the school solution.

$$\mu = p = 0.23, \qquad \sigma^2 = \frac{p(1-p)}{n} = \frac{0.23\times 0.77}{60} \doteq 0.0029517, \qquad \sigma \doteq 0.054329.$$

$$z = \frac{0.3-0.23}{0.054329} \doteq 1.2884 \doteq 1.29$$

From the extract, $P(Z < 1.29) = 0.9015$, so

$$P(\hat{p} > 0.3) \doteq P(Z > 1.29) = 1 - 0.9015 = 0.0985.$$

PP 5 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q14(c)**

(i) The mean is $\mu_{\hat{p}} = p = 0.7$ (not 0.67, which is the sample result). The standard deviation is

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.7\times 0.3}{1012}} \doteq 0.0144052.$$

(ii) Using the normal approximation to $\hat{p}$ without a continuity correction, as in the school solution.

The sample proportion is $\hat{p} = 0.67$. Here $np = 1012\times 0.7 = 708.4$ and $n(1-p) = 1012\times 0.3 = 303.6$, both greater than 5 (indeed greater than 10), so the normal approximation can be applied.

$$\begin{aligned} P(\hat{p} \le 0.67) &\doteq P\left(Z \le \frac{0.67-0.7}{0.0144052}\right) \\ &\doteq P(Z \le -2.08) \\ &= 1 - 0.9812 \\ &= 0.0188 \doteq 1.9\%. \end{aligned}$$

This probability is less than 0.05. If 70% of adults really drink the cereal milk, a poll result of 67% or less would be very unlikely, so there is doubt about the result of this poll.

PP 6 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q14(c)(i)**

By the empirical rule, 99.7% certainty means within 3 standard deviations either side of the mean p. So the maximum error of 0.001 must be at least 3 standard deviations:

$$3\sigma \leq 0.001, \qquad \text{so} \qquad \sigma \leq \frac{0.001}{3} = \frac{1}{3000}.$$

Using the estimate $p = 0.81$,

$$\sqrt{\frac{0.81 \times 0.19}{n}} \leq \frac{1}{3000}, \qquad \text{so} \qquad \frac{0.1539}{n} \leq \frac{1}{9\,000\,000}, \qquad \text{so} \qquad n \geq 0.1539 \times 9\,000\,000 = 1\,385\,100.$$

The mathematician needs at least 1 385 100 trials.

Chapter 17 Binomial Distributions - Homework Answers

Chapter 17 Review | Chapter review exercise

1 **a** ${}^4C_3\left(\frac{5}{6}\right)^3\left(\frac{1}{6}\right) = \frac{125}{324} \doteq 0.386$ **b** ${}^4C_2\left(\frac{5}{6}\right)^2\left(\frac{1}{6}\right)^2 = \frac{25}{216} \doteq 0.116$

2 $\left(\frac{5}{6}\right)^{15} + {}^{15}C_1\left(\frac{5}{6}\right)^{14}\left(\frac{1}{6}\right) + {}^{15}C_2\left(\frac{5}{6}\right)^{13}\left(\frac{1}{6}\right)^2 \doteq 0.532$

3 ${}^{10}C_7(0.2)^7(0.8)^3 \doteq 0.000786$

4 We need $1 - \left(\frac{3}{4}\right)^n > 0.95$. Thus $n > \dfrac{\log 0.05}{\log 0.75} \doteq 10.4$. The experiment needs to be repeated 11 times.

5 **a** $\frac{1}{8}$
b $X \sim B\left(6, \frac{1}{8}\right)$:

x	0	1	2	3	4	5	6
$P(X = x)$	0.4488	0.3847	0.1374	0.0262	0.0028	0.0002	0.0000

c $1000 \times 0.0262 \doteq 26$ times **d** $1000 \times (0.0262 + 0.0028 + 0.0002) \doteq 29$ times

6 **a** Yes. $p = 0.5$, $q = 0.5$ **b** Yes, if success is interpreted as the player winning. $p = \frac{1}{12}$ and $q = \frac{11}{12}$.
c Yes, with experimental probability $p = 0.004$, $q = 0.996$ **d** No, because a Bernoulli trial must have only two possible outcomes.

7 Mean, variance, standard deviation:
a 4, 3.2, 1.79 **b** 35, 17.5, 4.18 **c** 4.8, 0.96, 0.98 **d** 48, 28.8, 5.37 **e** 30, 27, 5.20 **f** 1.25, 0.9375, 0.97

8 **a** $X \sim B(60, 0.05)$: mean 3, variance 2.85, standard deviation $\doteq 1.69$
b $3 - 1.69 \le X \le 3 + 1.69$ gives $X = 2$, 3 or 4. $P(X = 2,\ 3 \text{ or } 4) \doteq 62.8\%$
c $P(0 \le X \le 4) \doteq 82\%$. The probability of rejection is 18%.
d Mean 1.2, standard deviation $\doteq 1.08$, so $X \le 2.28$. $P(X = 0,\ 1 \text{ or } 2) \doteq 88\%$. The probability of rejection is 12%.

9 **a** $\left(\frac{1}{2}\right)^{80}\left({}^{80}C_{38} + {}^{80}C_{39} + {}^{80}C_{40}\right) \doteq 25.62\%$ **b** 40, $\sqrt{20} \doteq 4.47$
c $np = nq = 40 > 5$, so a normal approximation may be used.
d This is called a continuity correction, and occurs because we are approximating a discrete distribution by a continuous curve. The columns for 38, 39 and 40 run from 37.5 to 40.5.
e The z-scores are -0.56 and 0.11, so $P(37.5 \le X \le 40.5) \doteq 0.5438 - 0.2877 = 0.2561 = 25.61\%$.
f $\dfrac{25.62 - 25.61}{25.62} \times 100\% \doteq 0.04\%$, which is less than 0.1%
g $P(X \ge 49.5) \doteq P(Z \ge 2.12) = 1 - 0.9830 = 0.0170 \doteq 1.7\%$

10 **a** $\frac{6}{36} = \frac{1}{6}$ **b** If $X \sim B\left(80, \frac{1}{6}\right)$, then $\mu = \frac{40}{3}$, $\sigma = \frac{10}{3}$, and
$P(X \ge 15) \doteq P(X \ge 14.5) \doteq P(Z \ge 0.35) = 1 - 0.6368 = 0.3632 \doteq 36\%$.

11 **a** 0.35 **b** Not necessarily. The number will vary from sample to sample.
c The number of yes answers has a binomial distribution with $n = 100$ and p equal to the (unknown) proportion of the population intending to vote for the WTP. The sample proportions have the corresponding sample proportion distribution, with mean p, and it is approximately normal.

12 **a**

$\hat{p}$	0	0.2	0.4	0.6	0.8	1
$P(\hat{p})$	$\frac{32}{3125}$	$\frac{48}{625}$	$\frac{144}{625}$	$\frac{216}{625}$	$\frac{162}{625}$	$\frac{243}{3125}$

b (i) $\frac{272}{3125}$ (ii) $\frac{992}{3125}$
c Mean $\mu_{\hat{p}} = \frac{3}{5}$, variance $\frac{6}{125}$, $\sigma_{\hat{p}} = \frac{1}{25}\sqrt{30} \doteq 0.219$

13 **a** $\mu = p = \frac{1}{6}$, $\sigma = \sqrt{\frac{1}{6} \times \frac{5}{6} \div 500} = \frac{1}{60}$
b The z-score is $\left(\dfrac{70}{500} - \dfrac{1}{6}\right) \div \dfrac{1}{60} = -1.6$, so the result is 1.6 standard deviations below the mean.

14 $\mu = 0.53$, $\sigma = \sqrt{0.53 \times 0.47 \div 653} \doteq 0.01953$. $P(\hat{p} > 0.54) \doteq P(Z > 0.51) = 1 - 0.6950 = 0.3050 \doteq 30.5\%$